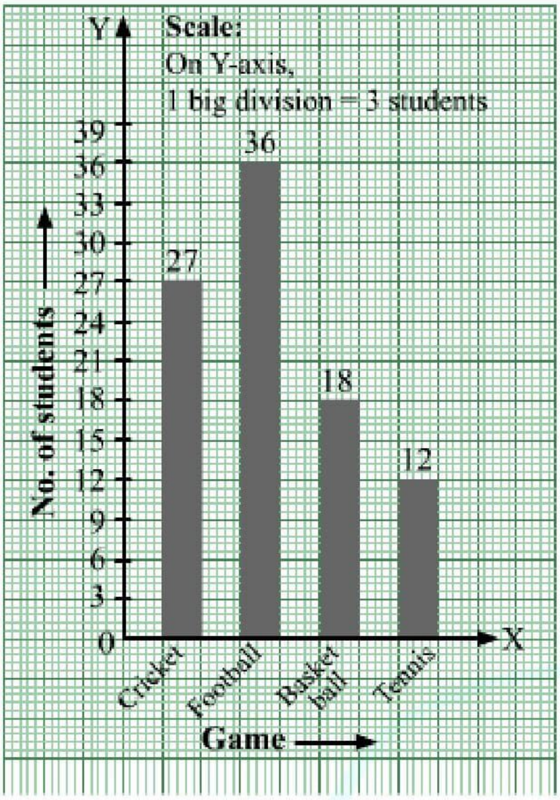
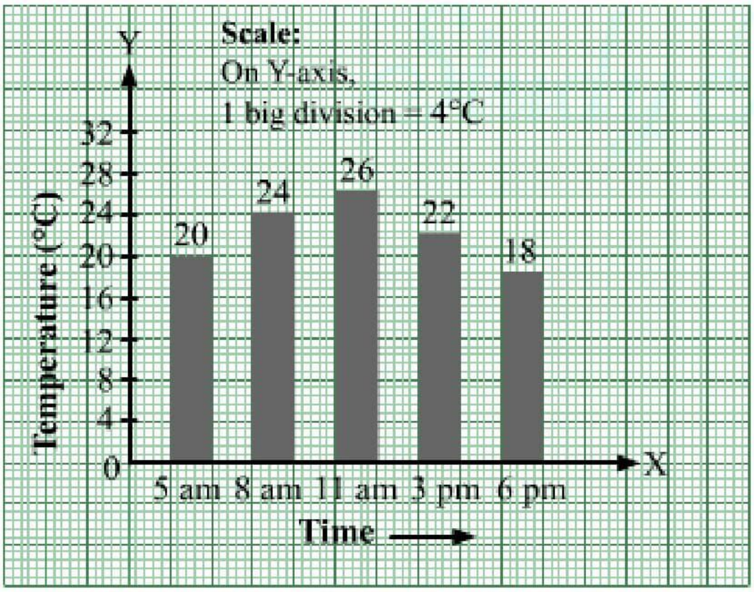
Exercise 17A

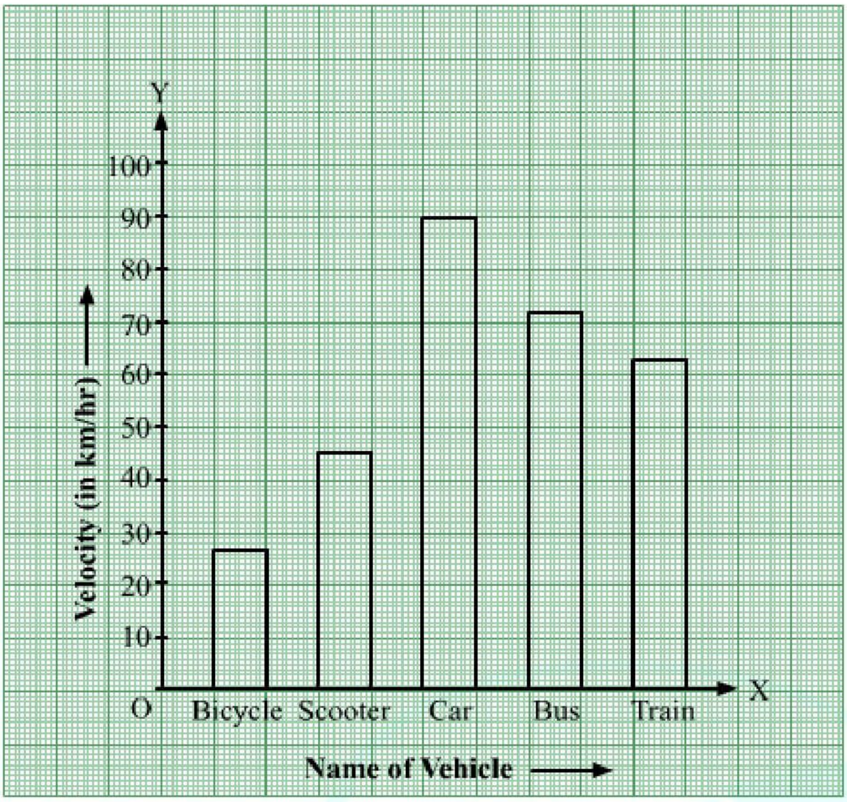
Solution 1:



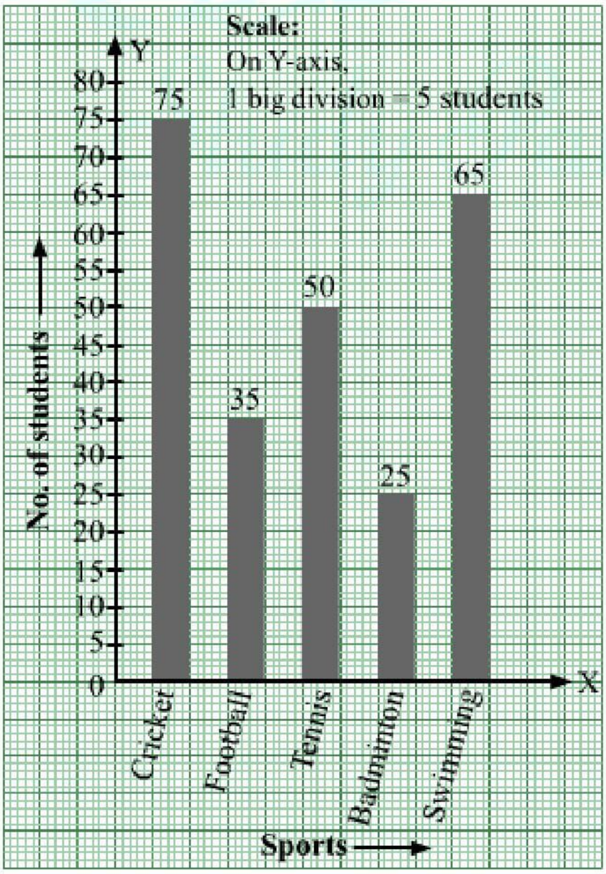
Solution 2:



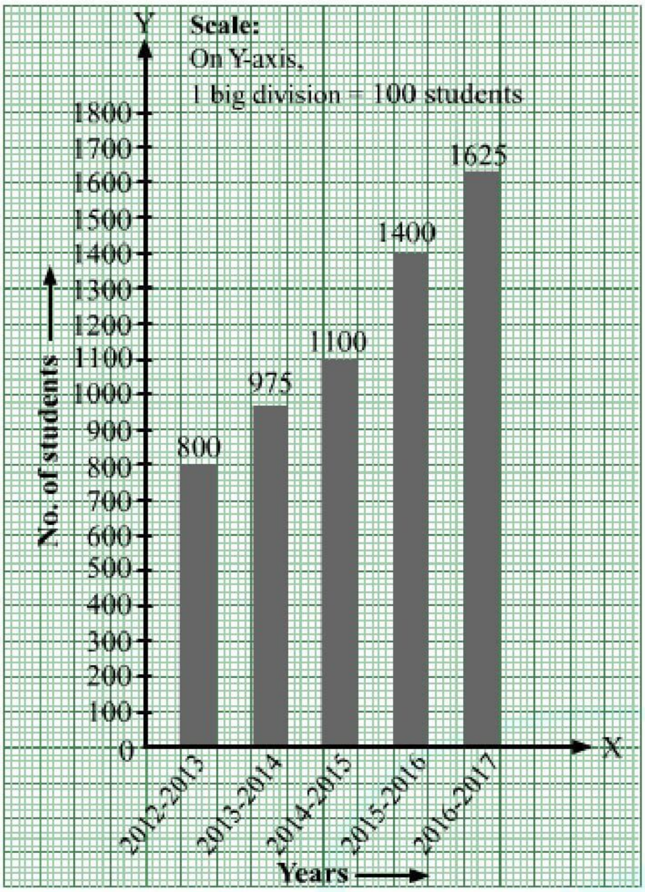
Solution 3: Take the name of vehicle along the *x*-axis and the velocity along the *y*-axis.



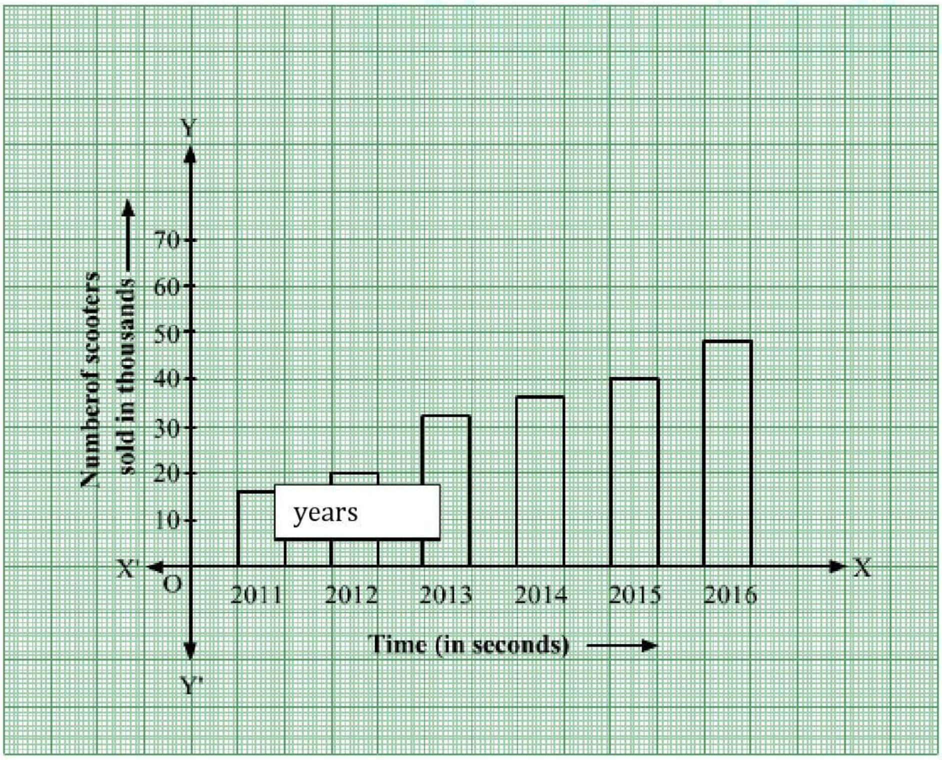
Solution 4:



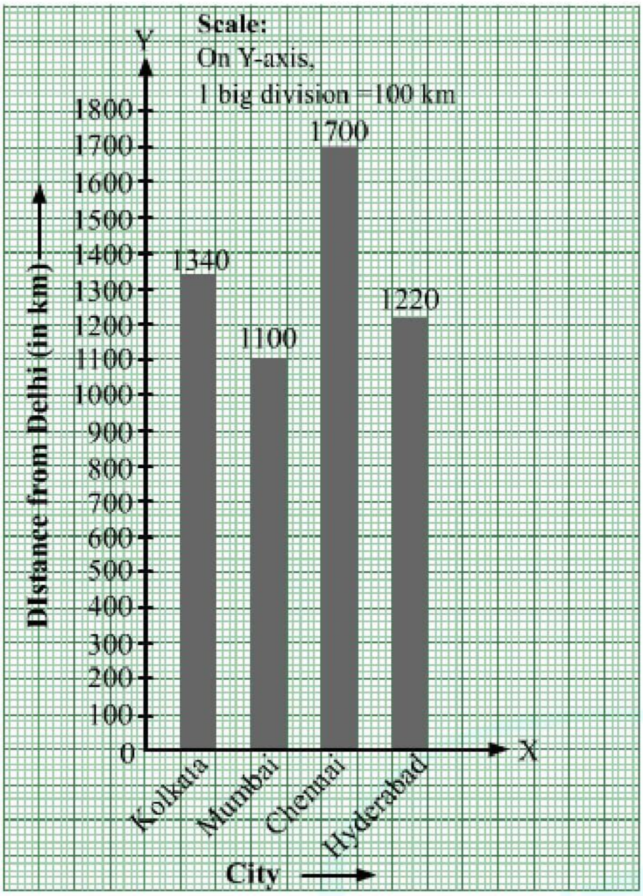
Solution 5:



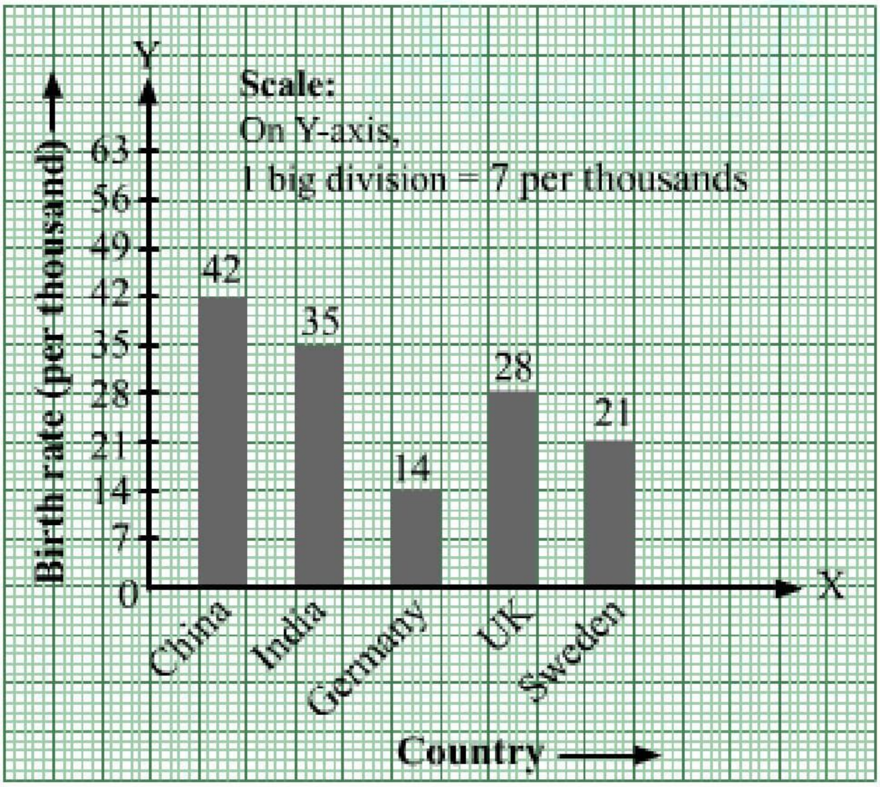
Solution 6:



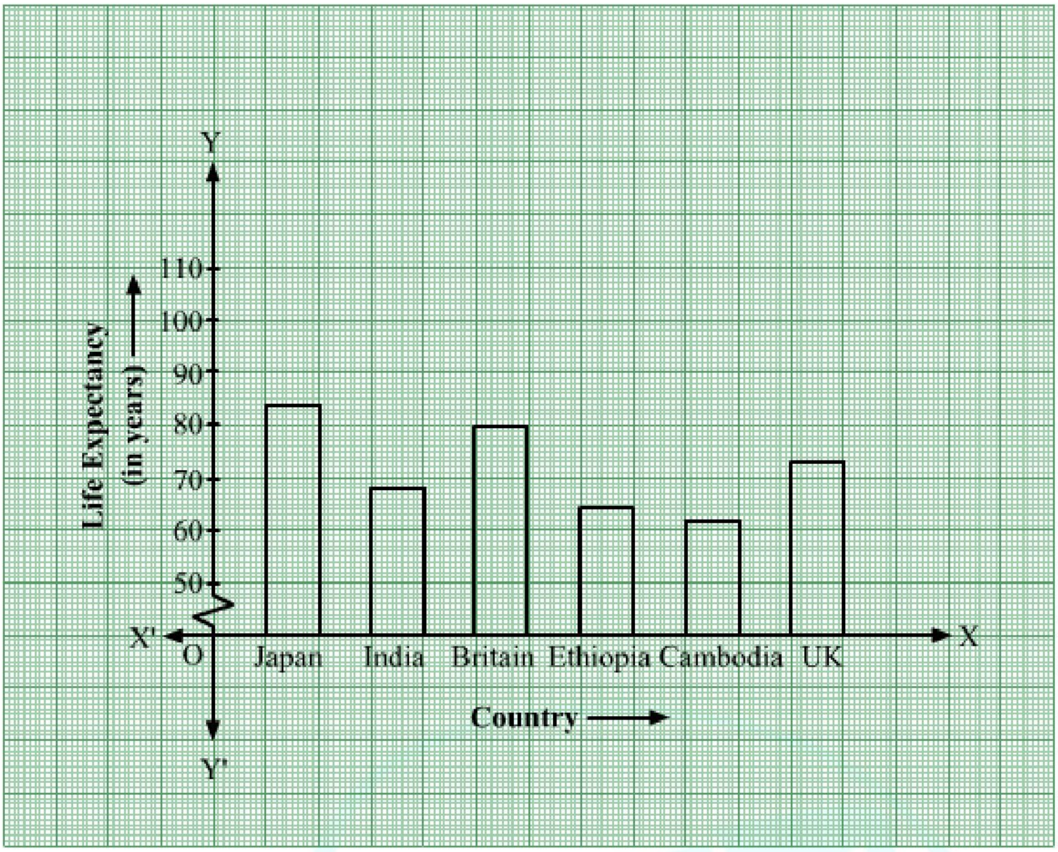
Solution 7:



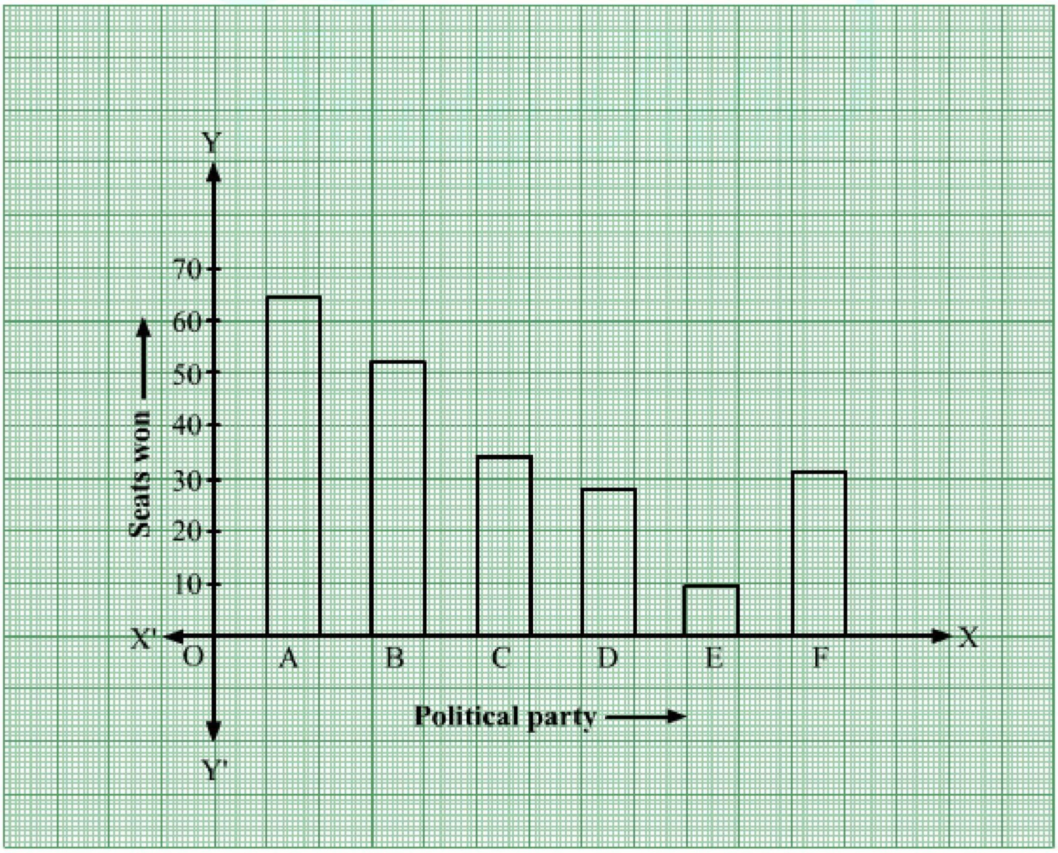
Solution 8:



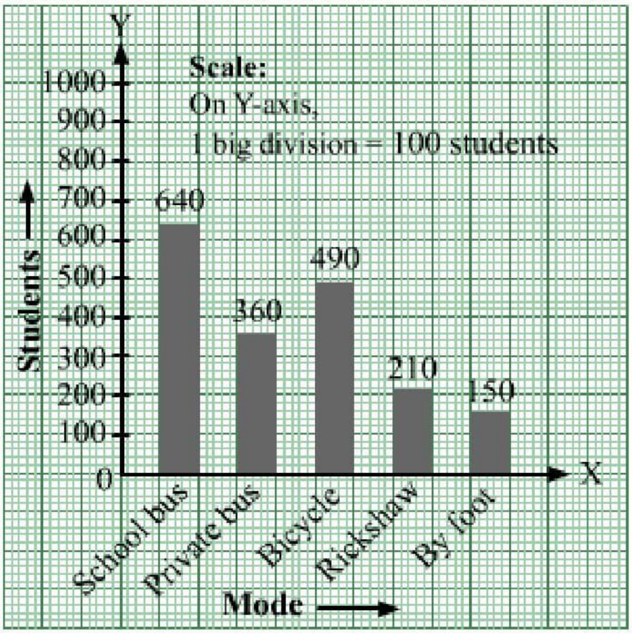
Solution 9:



Solution 10:



Solution 11:



Solution 12: (i) The bar graph shows the marks obtained by a student in various subjects in an examination.

(ii) The student scores very good in mathematics, as the height of the corresponding bar is the highest.

(iii) The student scores bad in Hindi, as the height of the corresponding bar is the lowest.

(iv) Average marks $= \dfrac{60 + 35 + 75 + 50 + 60}{5} = \dfrac{280}{5} = 56$

Exercise 17B

Solution 1: The given frequency distribution is in exclusive form.

We will represent the class intervals [daily wages (in rupees)] along the x-axis & the corresponding frequencies [number of workers] along the y-axis.

The scale is as follows:

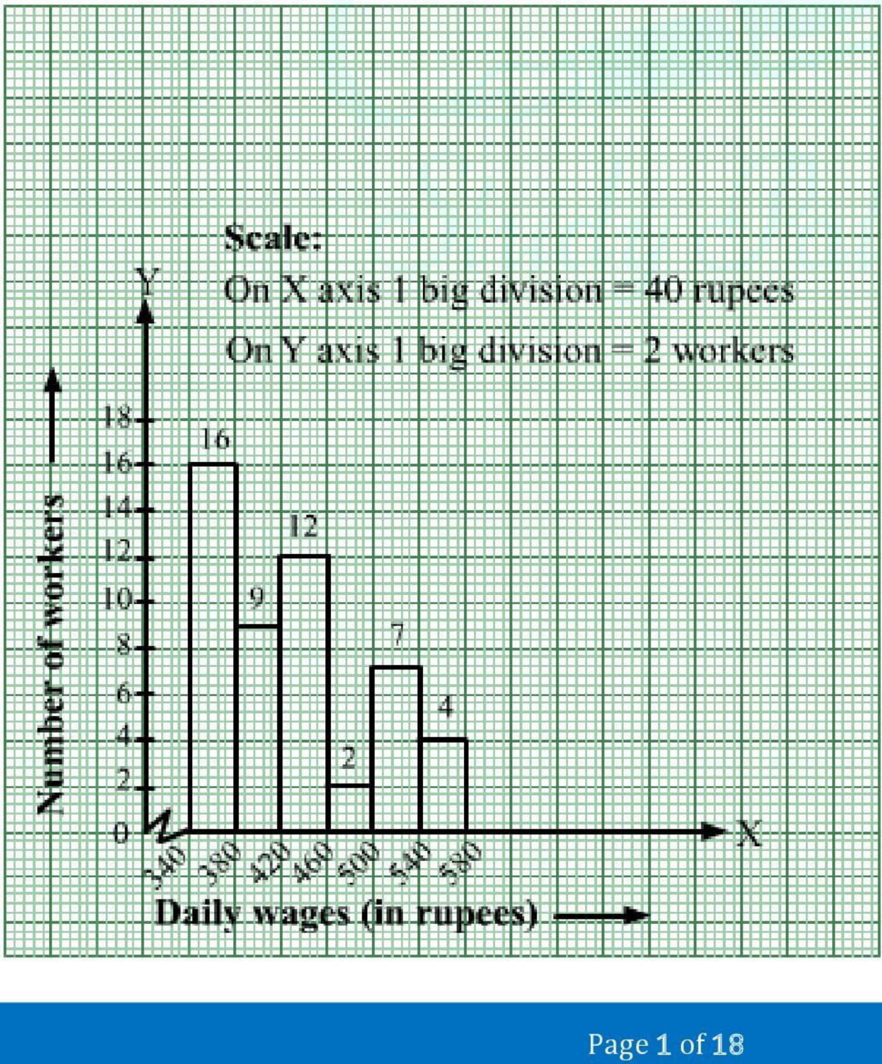
On x-axis: 1 big division = 40 rupees

On y-axis: 1 big division = 2 workers

Because the scale on the x-axis starts at 340, a kink, i.e., a break, is indicated near the origin to signify that the graph is drawn with a scale beginning at 340 and not at the origin.

We will construct rectangles with the class intervals as bases and the corresponding frequencies as heights.

Thus, we will obtain the following histogram:



Solution 2: The given frequency distribution is in exclusive form.

We will represent the class intervals [daily earnings (in rupees)] along the x-axis & the corresponding frequencies [number of stores] along the y-axis.

The scale is as follows:

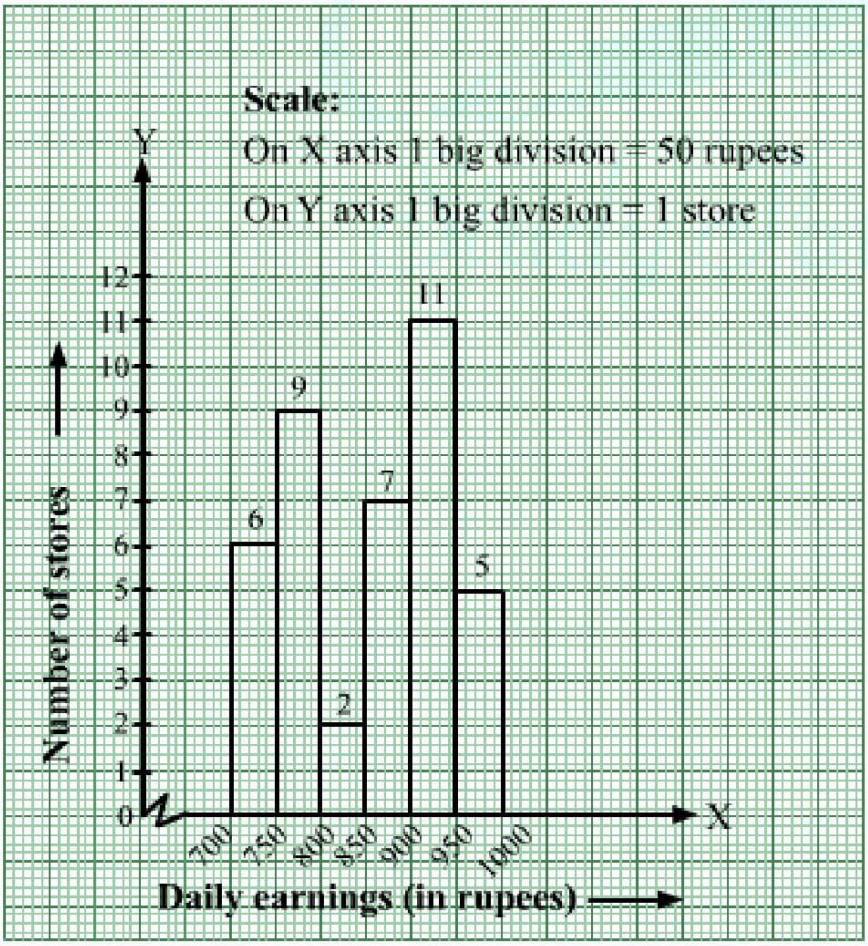
On x-axis: 1 big division = 50 rupees

On y-axis: 1 big division = 1 store

Because the scale on the x-axis starts at 700, a kink, i.e., a break is indicated near the origin to signify that the graph is drawn with a scale beginning at 700 and not at the origin.

We will construct rectangles with the class intervals as bases and the corresponding frequencies as heights.

Thus, we will obtain the following histogram:



Solution 3: The given frequency distribution is in exclusive form.

We will represent the class intervals [heights (in cm)] along the x-axis & the corresponding frequencies [number of students] along the y-axis.

The scale is as follows:

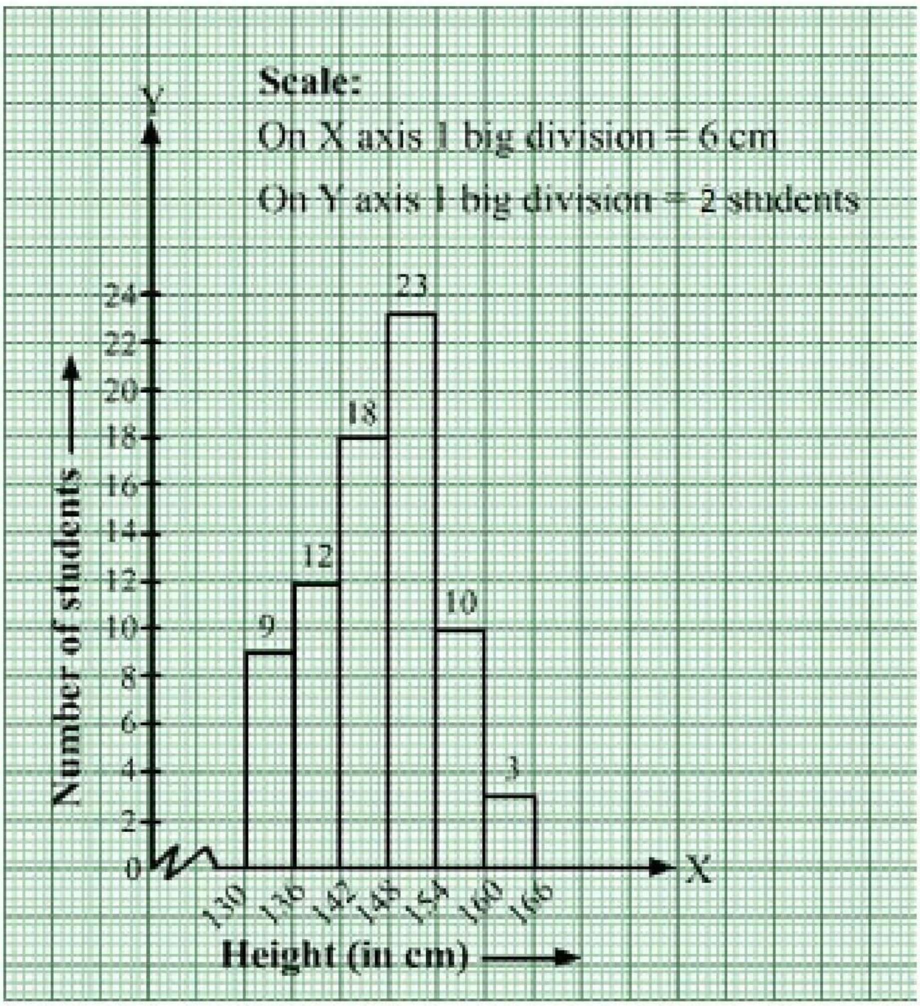
On x-axis: 1 big division = 6 cm

On y-axis: 1 big division = 2 students

Because the scale on the x-axis starts at 130, a kink, i.e., a break, is indicated near the origin to signify that the graph is drawn with a scale beginning at 130 and not at the origin.

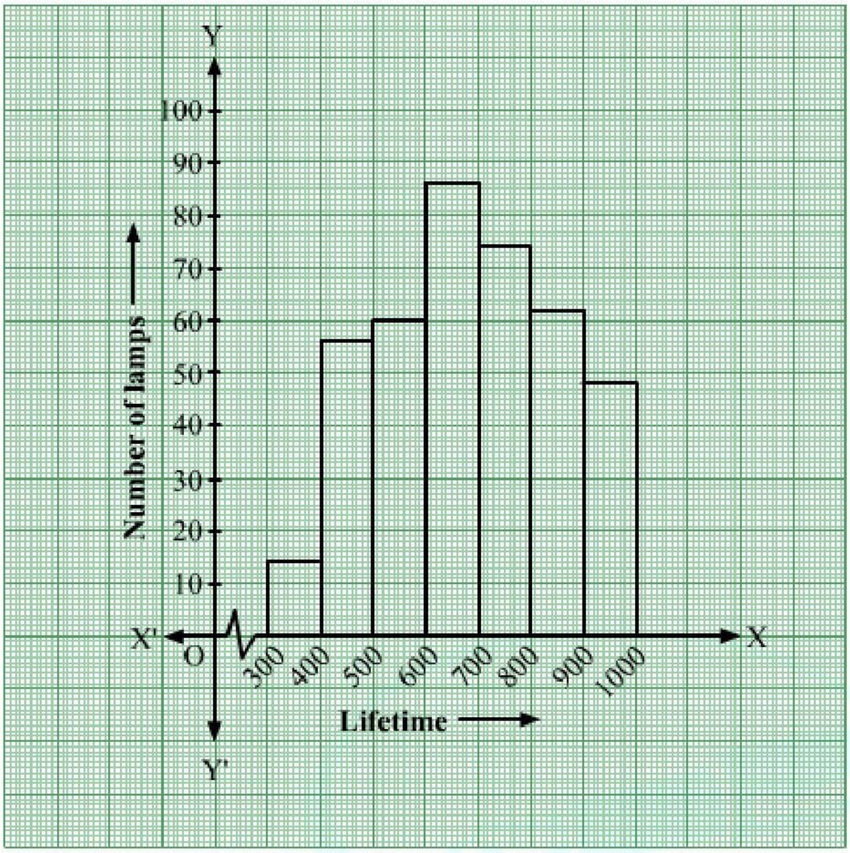
We will construct rectangles with the class intervals as bases and the corresponding frequencies as heights.

Thus, we will obtain the following histogram:



Solution 4:

(i)



(ii) Lamps with lifetime more than 700 hours $= 74 + 62 + 48 = 184$

Solution 5: The given frequency distribution is in exclusive form.

We will represent the class intervals along the x-axis & the corresponding frequencies along the y-axis.

The scale is as follows:

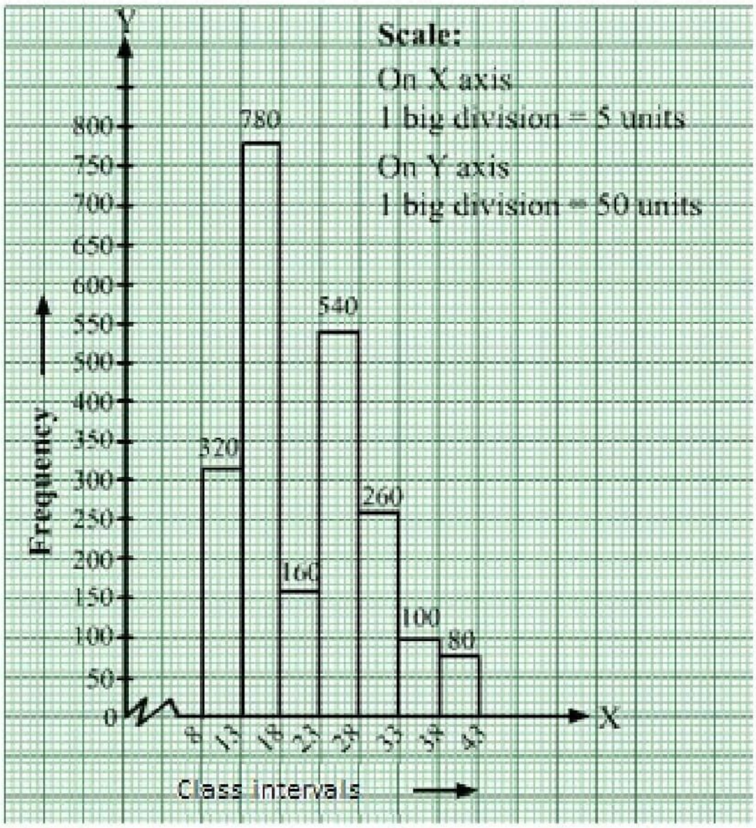
On x-axis: 1 big division $= 5$ units

On y-axis: 1 big division $= 50$ units

Because the scale on the x-axis starts at 8, a kink, i.e., a break, is indicated near the origin to signify that the graph is drawn with a scale beginning at 8 and not at the origin.

We will construct rectangles with the class intervals as bases and the corresponding frequencies as heights.

Thus, we will obtain the following histogram:



Solution 6: The given frequency distribution is in inclusive form.

So, we will convert it into exclusive form, as shown below:

Class Interval	Frequency
4.5–12.5	6
12.5–20.5	15
20.5–28.5	24
28.5–36.5	18
36.5–44.5	4
44.5–52.5	9

We will mark class intervals along the x-axis and frequencies along the y-axis.

The scale is as follows:

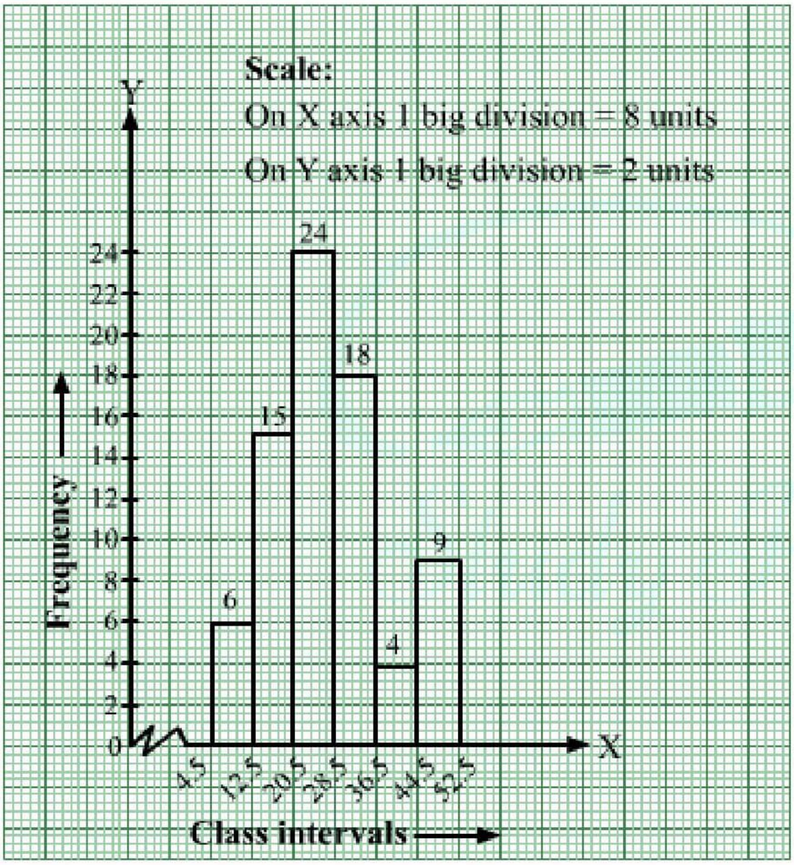
On x-axis: 1 big division = 8 units

On y-axis: 1 big division = 2 units

Because the scale on the x-axis starts at 4.5, a kink, i.e., a break, is indicated near the origin to signify that the graph is drawn with a scale beginning at 4.5 and not at the origin.

We will construct rectangles with class intervals as bases and the corresponding frequencies as heights.

Thus, we will obtain the histogram as shown below:



Solution 7: The given frequency distribution is inclusive form.

So, we will convert it into exclusive form, as shown below:

Age (in years)	Number of Illiterate Persons
9.5-16.5	175
16.5-23.5	325

23.5-30.5	100
30.5-37.5	150
37.5-44.5	250
44.5-51.5	400
51.5-58.5	525

We will mark the age groups (in years) along the x-axis & frequencies (number of illiterate persons) along the y-axis.

The scale is as follows:

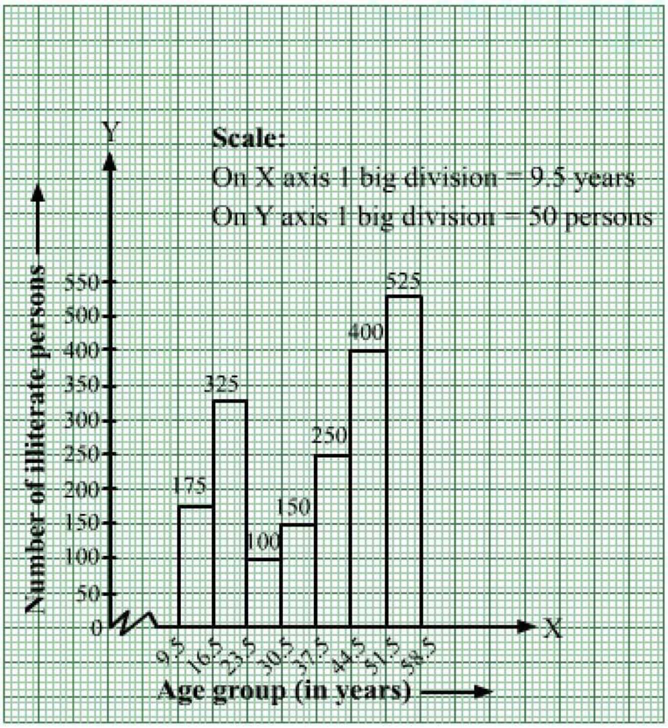
On x-axis: 1 big division = 7 years

On y-axis: 1 big division = 50 persons

Because the scale on the x-axis starts at 9.5, a kink, i.e., a break, is indicated near the origin to signify that the graph is drawn with a scale beginning at 9.5 and not at the origin.

We will construct rectangles with class intervals (age) as bases and the corresponding frequencies (number of illiterate persons) as heights.

Thus, we obtain the histogram, as shown below:



Solution 8:

In the given frequency distribution, class sizes are different.

So, we calculate the adjusted frequency for each class.

The minimum class size is 4.

$$\text{Adjusted frequency of a class } = \frac{\text{Minimum class size}}{\text{Class size of the class}} \times \text{Its frequency}$$

We have the following table:

Class Interval	Frequency	Adjusted Frequency
10-14	5	$\frac{4}{4} \times 5 = 5$
14-20	6	$\frac{4}{6} \times 6 = 4$
20-32	9	$\frac{4}{12} \times 9 = 3$
32-52	25	$\frac{4}{20} \times 25 = 5$
52-80	21	$\frac{4}{28} \times 21 = 3$

We mark the class intervals along the *x*-axis and the corresponding adjusted frequencies along the *y*-axis.

We have chosen the scale as follows:

On the *x*- axis,

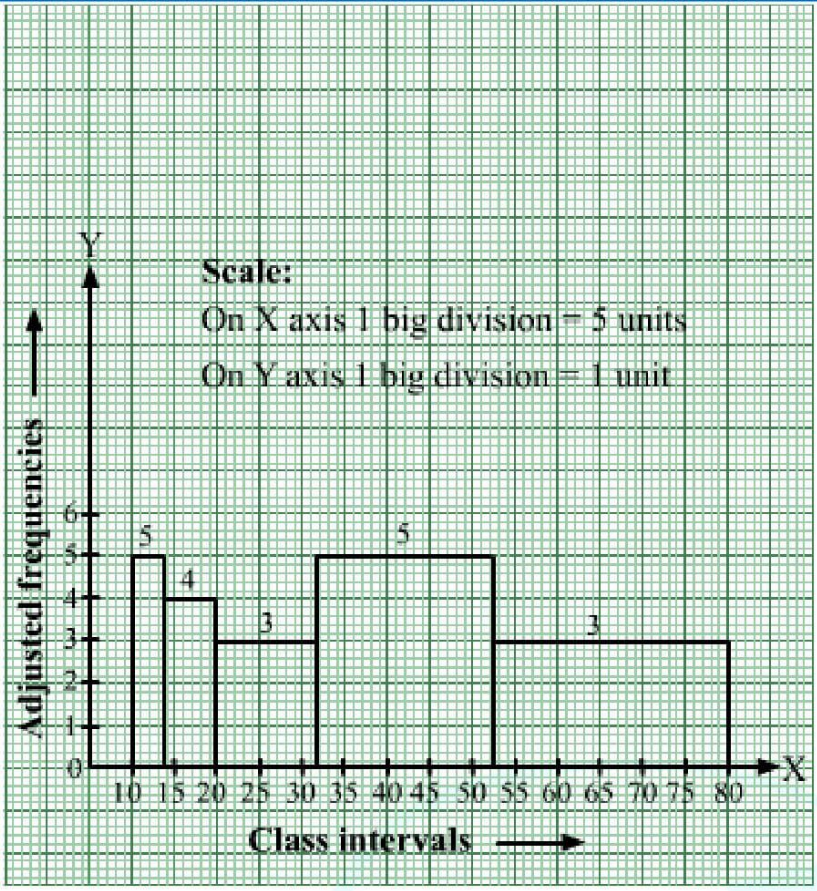
1 big division = 5 units

On the *y*-axis,

1 big division = 1 unit

We draw rectangles with class intervals as the bases and the corresponding adjusted frequencies as the heights.

Thus, we obtain the following histogram:



Solution 9:

(i) Here the class intervals are of unequal size. So, we calculate the adjusted frequency using the formula,

$$\text{Adjusted frequency} = \frac{\text{min class size}}{\text{class size of this class}} \times \text{frequency}$$

Class sizes are

$4 - 1 = 3$

$6 - 4 = 2$

$8 - 6 = 2$

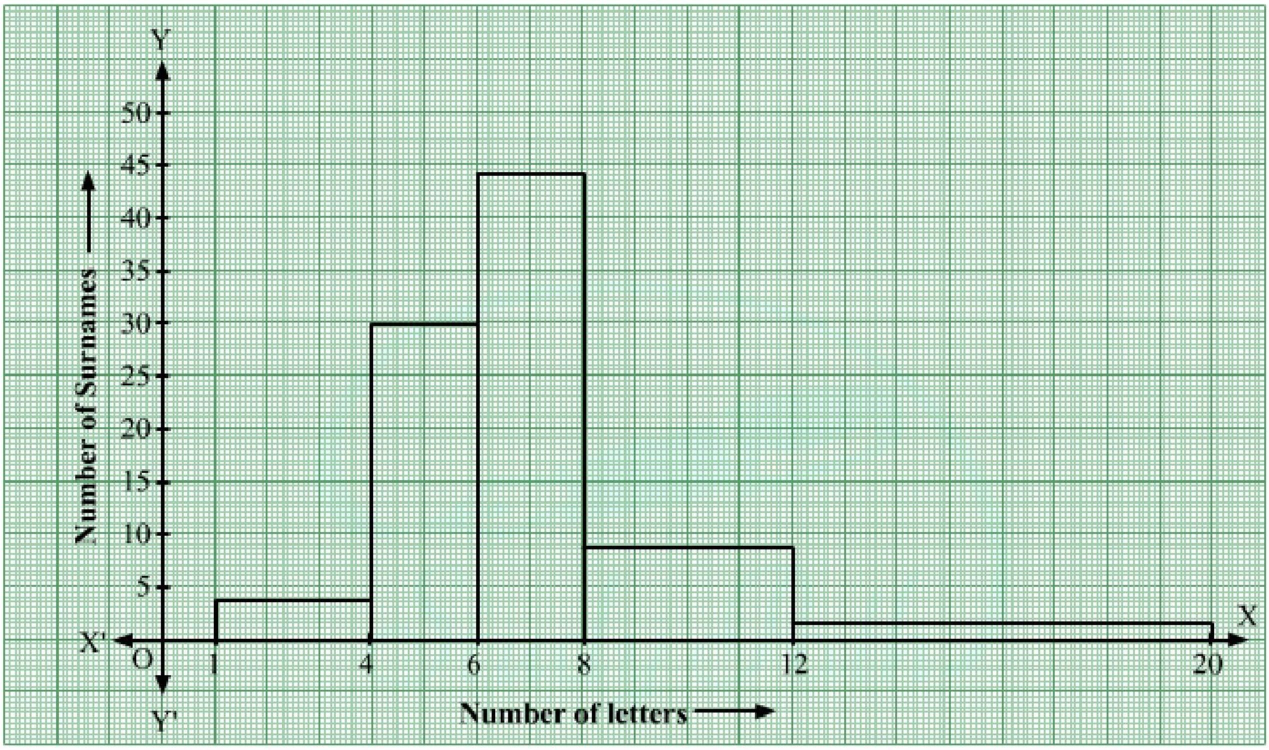
$12 - 8 = 4$

$20 - 12 = 8$

Minimum class size $= 2$

Number of letters	Frequency	Width of class	Height of rectangle
1-4	6	3	$\frac{2}{3} \times 6 = 4$

4-6	30	2	$\frac{2}{2}\times 30 = 30$
6-8	44	2	$\frac{2}{2}\times 44 = 44$
8-12	16	4	$\frac{2}{4}\times 16 = 8$
12-20	4	8	$\frac{2}{8}\times 4 = 1$



(ii) Maximum number of surnames lie in the interval 6-8.

Solution 10: Here the class intervals are of unequal size. So, we calculate the adjusted frequency using the formula,

Adjusted frequency $= \frac{\text{min class size}}{\text{class size of this class}} \times \text{frequency}$

Class sizes are

$10 - 5 = 5$

$15 - 10 = 5$

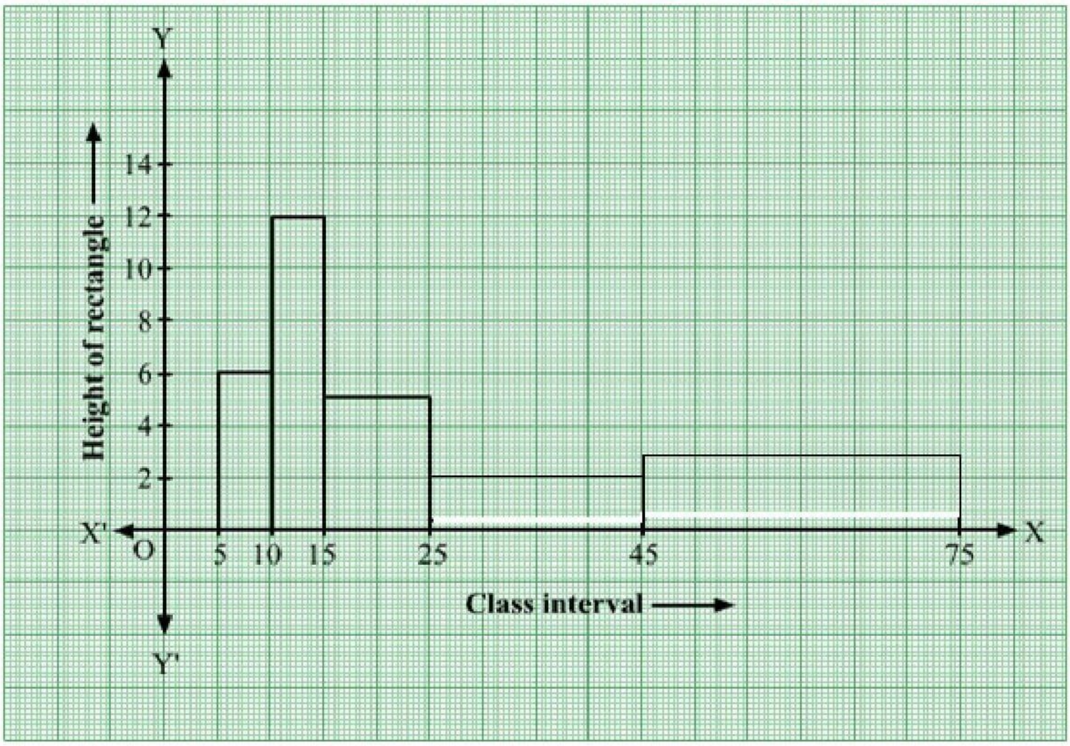
$25 - 15 = 10$

$45 - 25 = 20$

$75 - 45 = 30$

Minimum class size $= 5$

Class interval	Frequency	Width of class	Height of rectangle
5-10	6	5	$\frac{5}{5} \times 6 = 6$
10-15	12	5	$\frac{5}{5} \times 12 = 12$
15-25	10	10	$\frac{5}{10} \times 10 = 5$
25-45	8	20	$\frac{5}{20} \times 8 = 2$
45-75	18	30	$\frac{5}{30} \times 18 = 3$

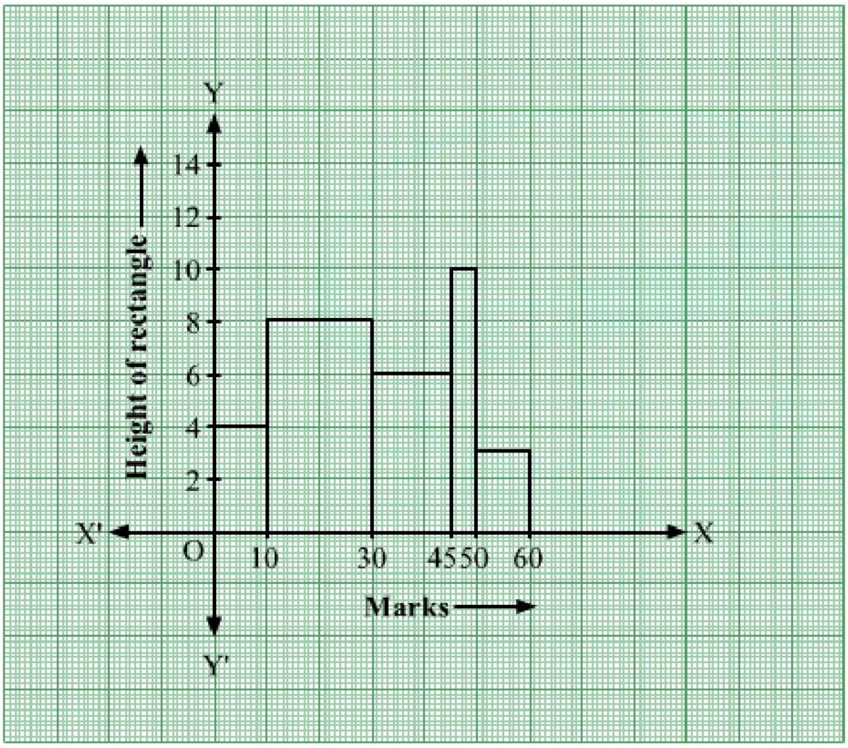


Solution 11: Here the class intervals are of unequal size. So, we calculate the adjusted frequency using the formula,

$$\text{Adjusted frequency} = \frac{\text{min class size}}{\text{class size of this class}} \times \text{frequency}$$

Minimum class size $= 5$

Marks	Frequency	Width of class	Height of rectangle
0-10	8	10	$\frac{5}{10} \times 8 = 4$
10-30	32	20	$\frac{5}{20} \times 32 = 8$
30-45	18	15	$\frac{5}{15} \times 18 = 6$
45-50	10	5	$\frac{5}{5} \times 10 = 10$
50-60	6	10	$\frac{5}{10} \times 6 = 3$



Solution 12:

We take two imagined classes—one at the beginning (0–10) and other at the end (70–80)—each with frequency zero.

With these two classes, we have the following frequency table:

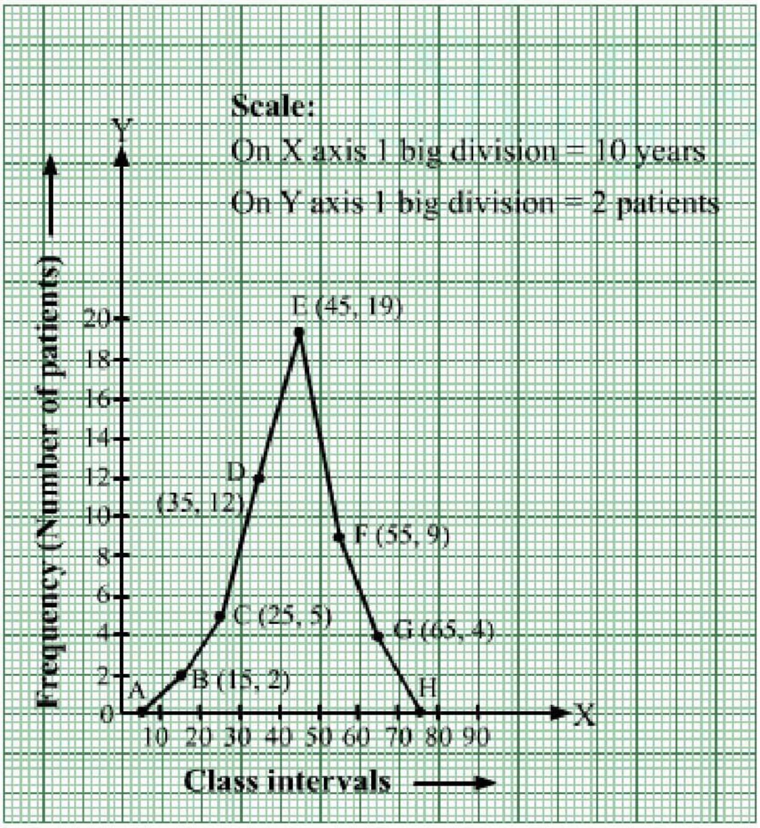
Class mark of the interval $= \dfrac{\text{upper limit} + \text{lower limit}}{2}$

Age in Years	Class Mark	Frequency (Number of Patients)
0–10	5	0
10–20	15	2
20–30	25	5
30–40	35	12
40–50	45	19
50–60	55	9
60–70	65	4
70–80	75	0

Now, we plot the following points on a graph paper:

A(5, 0), B(15, 2), C(25, 5), D(35, 12), E(45, 19), F(55, 9), G(65, 4) and H(75, 0)

Join these points with line segments AB, BC, CD, DE, EF, FG, GH ,HI and IJ to obtain the required frequency polygon.



Solution 13: Though the given frequency table is in inclusive form, class marks in case of inclusive and exclusive forms are the same.

We take the imagined classes (-9)–0 at the beginning and 61–70 at the end, each with frequency zero.

Class mark of the interval $= \dfrac{\text{upper limit} + \text{lower limit}}{2}$

Thus, we have:

Class Interval	Class Mark	Frequency
−9–0	–4.5	0
1–10	5.5	8
11–20	15.5	3
21–30	25.5	6
31–40	35.5	12
41–50	45.5	2
51–60	55.5	7
61–70	65.5	0

Along the x-axis, we mark –4.5, 5.5, 15.5, 25.5, 35.5, 45.5, 55.5 and 65.5.

Along the y-axis, we mark 0, 8, 3, 6, 12, 2, 7 and 0.

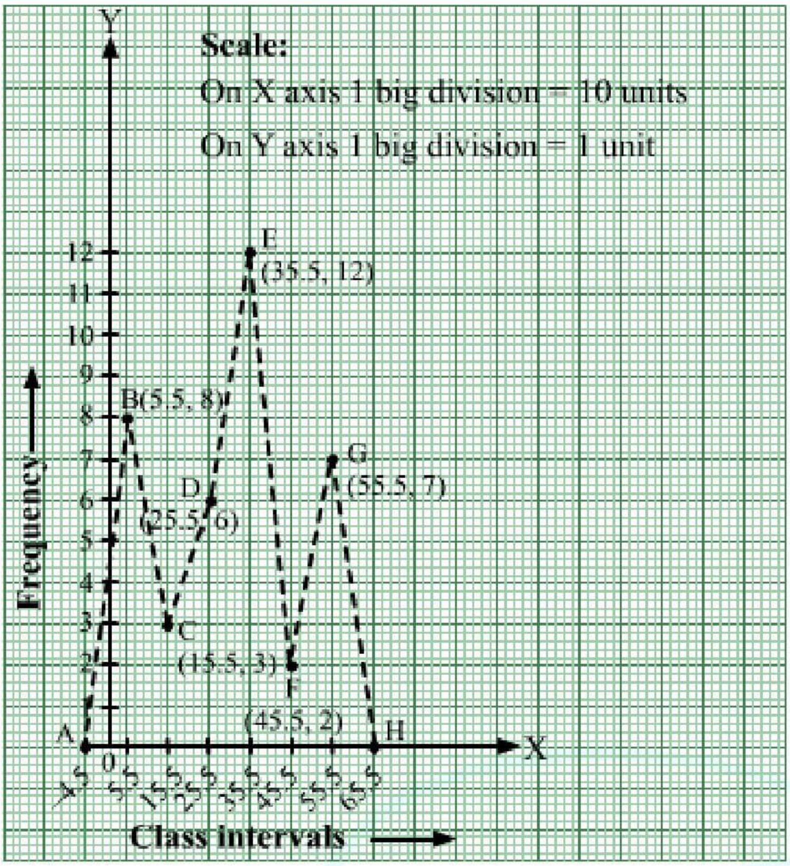
We have chosen the scale as follows:

On the x-axis, 1 big division = 10 units.

On the y-axis, 1 big division = 1 unit.

We plot the points A(–4.5,0), B(5.5, 8), C(15.5, 3), D(25.5, 6), E(35.5, 12), F(45.5, 2), G(55.5, 7) and H(65.5, 0).

We draw line segments AB, BC, CD, DE, EF, FG, GH to obtain the required frequency polygon, as shown below.



Solution 14: We represent the class intervals along the x-axis and the corresponding frequencies along the y-axis.

We construct rectangles with class intervals as bases and respective frequencies as heights.

We have the scale as follows:

On the x-axis:

1 big division = 10 years

On the y-axis:

1 big division = 2 patients

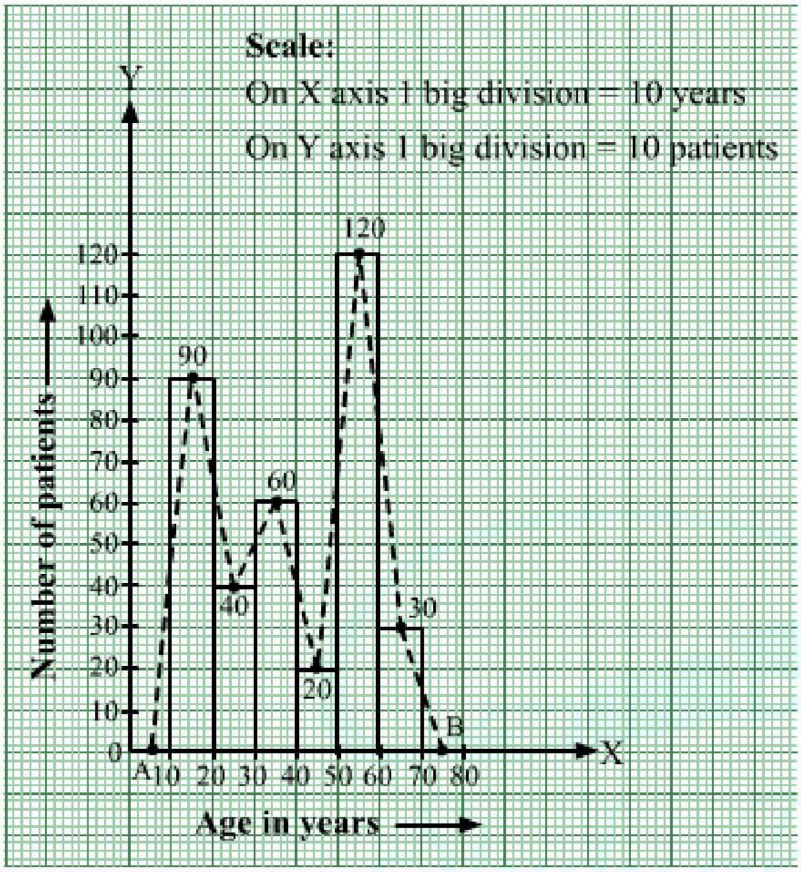
Thus, we obtain the histogram, as shown below.

We join the midpoints of the tops of adjacent rectangles by line segments.

Also, we take the imagined classes 0–10 and 70–80, each with frequency 0. The class marks of these classes are 5 and 75, respectively.

So, we plot the points A(5, 0) and B(75, 0). We join A with the midpoint of the top of the first rectangle and B with the midpoint of the top of the last rectangle.

Thus, we obtain a complete frequency polygon, as shown below:



Solution 15: We represent the class intervals along the x-axis and the corresponding frequencies along the y-axis.

We construct rectangles with class intervals as bases and respective frequencies as heights.

We have the scale as follows:

On the x-axis, 1 big division = 5 units.

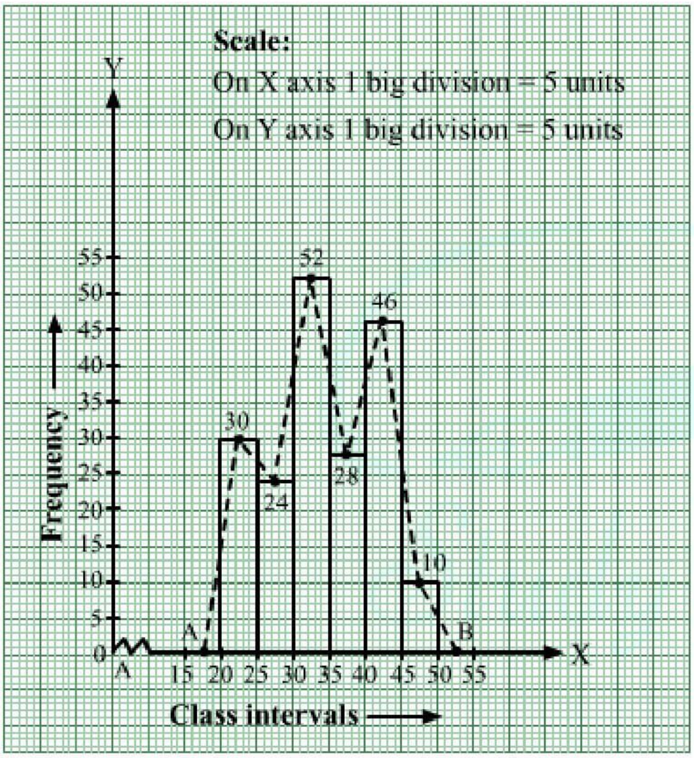
On the y-axis, 1 big division = 5 units.

Because the scale on the x-axis starts at 15, a kink, i.e., a break, is indicated near the origin to signify that the graph is drawn with a scale beginning at 15
and not at the origin.

Thus, we obtain the histogram, as shown below.

We join the midpoints of the tops of adjacent rectangles by line segments.
Also, we take the imagined classes 15–20 and 50–55, each with frequency 0. The class marks of these classes are 17.5 and 52.5, respectively.

So, we plot the points $A(17.5, 0)$ and $B(52.5, 0)$. We join A with the midpoint of the top of the first rectangle and B with the midpoint of the top of the last rectangle.
Thus, we obtain a complete frequency polygon, as shown below:



Solution 16: We represent the class intervals along the x-axis and the corresponding frequencies along the y-axis.
We construct rectangles with class intervals as bases and respective frequencies as heights.
We have the scale as follows:
On the x-axis:
1 big division = 40 units

On the y-axis:

1 big division = 20 units

Thus, we obtain the histogram, as shown below:

We join the midpoints of the tops of adjacent rectangles by line segments.

Also, we take the imagined classes 560–600 and 840–880, each with frequency 0.

The class marks of these classes are 580 and 860, respectively.

Because the scale on the x-axis starts at 560, a kink; i.e., a break, is indicated near the origin to signify that the graph is drawn with a scale beginning at 560
and not at the origin.

So, we plot the points $A(580, 0)$ and $B(860, 0)$. We join A with the midpoint of the top of the first rectangle and join B with the midpoint of the top of the last rectangle.

Thus, we obtain a complete frequency polygon, as shown below: